

CBIA

Customer Behavior Intelligence & Analytics

Business Requirements Document

Document Title	CBIA — Customer Behavior Intelligence & Analytics · BRD
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Client	XYZ Pvt. Ltd.
Project Phase	Phase 0 — Requirements & Planning
Status	Final — All 9 Phases Complete
Classification	

1. Executive Summary

CBIA (Customer Behavior Intelligence & Analytics) is a comprehensive, enterprise-grade data analytics solution developed for XYZ Pvt. Ltd. The project delivers end-to-end customer intelligence through machine learning, SQL analytics, and an interactive 5-page Power BI dashboard.

The solution integrates RFM (Recency, Frequency, Monetary) scoring, K-Means clustering, Customer Lifetime Value (CLV) prediction, and a Churn Prediction model — providing XYZ Pvt. Ltd. with actionable, data-driven insights for strategic decision-making across all customer touchpoints.

1.1 Project Outcome Summary

Total Revenue	Total Customers	Churn Rate	Avg CLV
\$1,963,079	5,000	42.7%	\$1,035
Avg Order Value	Repeat Rate	Total CLV	Avg Rating
\$392.62	99.7%	\$5,176,565	3.67 / 5.0

2. Business Problem

XYZ Pvt. Ltd. was facing the following critical business challenges that necessitated this analytics project:

#	Challenge	Business Impact
1	35% annual customer churn	Direct revenue loss and customer base erosion
2	Declining marketing ROI (-8% YoY)	Budget inefficiency with no targeting intelligence
3	No customer segmentation strategy	One-size-fits-all approach reduces conversion rates
4	No centralized analytics platform	Manual, siloed decisions without data support
5	No CLV visibility	Suboptimal resource allocation across customer tiers
6	Cannot identify At-Risk customers	High-value customer churn undetected until too late

3. Business Objectives & KPI Mapping

Business Objective	Target KPI	Measurement	Priority
Revenue Growth	+20%	Total Revenue	High
Churn Reduction	-15%	Churn Rate	Critical
Retention Improvement	+10%	Repeat Customers	High
AOV Increase	+12%	Average Order Value	Medium
CLV Optimization	+18%	Customer Lifetime Value	High

4. Project Scope

4.1 In Scope

- Customer segmentation using RFM scoring (Quintile 1–5) and K-Means clustering (5 clusters)
- CLV (Customer Lifetime Value) prediction model using Gradient Boosting Regressor
- Churn prediction model using Gradient Boosting Classifier (AUC-ROC target: 80%+)
- SQL analytics layer with 5 scripts — KPI queries, RFM layer, churn analytics, KPI summary view
- Power BI interactive dashboard — 5 pages with DAX measures, slicers, drill-through
- Win-back list generation for at-risk customers with recommended actions
- Python EDA, data cleaning, feature engineering across 5 Jupyter notebooks
- GitHub repository with complete documentation and README

4.2 Out of Scope

- Real-time streaming data pipelines or live data refresh
- Full AI production deployment and MLOps infrastructure
- Mobile application development
- CRM system integration or API endpoints

5. Data Architecture

The solution follows a structured 9-phase pipeline from raw data to business insights:

Phase	Stage	Description	Output File
1	Dataset	UCI-style synthetic data · 5,000 rows · 22 columns	customer_cleaned.csv
2	EDA & Cleaning	Null handling, outlier removal, Age_Group engineering	01_EDA_Data_Cleaning.ipynb
3	RFM Scoring	Quintile scoring R/F/M (1–5) · 6 segment labels	02_RFM_Scoring.ipynb + rfm_scored.csv
4	KMeans Clustering	K=5 clusters · PCA visualization · Silhouette scoring	03_KMeans_Clustering.ipynb + customers_segmented.csv
5	CLV Prediction	Gradient Boosting · 4-model comparison · CLV tiers	04_CLV_Prediction.ipynb + clv_predictions.csv
6	Churn Model	5-signal churn flag · AUC-ROC · Win-back list	05_Churn_Model.ipynb + churn_predictions.csv
7	SQL Analytics	5 scripts · KPI view · Window functions · MySQL	06_SQL_Scripts_MySQL.sql
8	Power BI	5-page dashboard · DAX measures · Slicers + drill-through	dashboard.pbix
9	PPT + GitHub	10-slide presentation + README + all deliverables	CBIA_Presentation_v2.pptx

5.1 Data Schema — customer_data Table

Column Name	Data Type	Description
Customer_ID	INT	Unique customer identifier
Age	INT	Customer age (18–70)
Age_Group	VARCHAR	Age bracket: 18-25, 26-35, 36-45, 46-55, 56-70
Gender	VARCHAR	Male / Female / Other
Location	VARCHAR	City / State
Order_Date	DATE	Date of purchase (Jan 2023 – Dec 2024)
Category	VARCHAR	Clothing / Accessories / Electronics / Footwear / Outerwear
Item_Purchased	VARCHAR	Product name
Quantity	INT	Units purchased

Revenue	DECIMAL	Final revenue after discount applied
Discount_Applied	VARCHAR	Yes / No — discount flag
Payment_Method	VARCHAR	Debit / Credit / Cash / PayPal / Venmo / Bank Transfer
Shipping_Type	VARCHAR	Standard / Express / Free / Pickup
Season	VARCHAR	Fall / Spring / Summer / Winter
Review_Rating	DECIMAL	Customer rating 1.0 – 5.0
Subscription_Status	VARCHAR	Yes / No — subscription membership
Previous_Purchases	INT	Historical purchase count (0–50)
Frequency_of_Purchases	VARCHAR	Weekly / Monthly / Quarterly / Annually etc.

6. Functional Requirements — Power BI Dashboard

6.1 Page 1 — Executive KPIs (Overview)

- 6 KPI Cards: Total Revenue \$1.96M · Total Customers 5,000 · AOV \$392.62 · Churn Rate 42.7% · Repeat Rate 99.7% · Avg CLV \$1,035
- Monthly Revenue Trend 2023 vs 2024 — Clustered column chart with YoY comparison
- Revenue by Category — Donut chart (Clothing 39.6% · Accessories 31.1% · Electronics 21.1%)
- Revenue by Season — Bar chart (Fall \$560K · Spring \$479K · Winter \$466K · Summer \$458K)
- KPI vs Target — 3 gauge charts (Revenue · Churn Rate · Retention Rate · AOV)
- Top 5 Customers by Revenue — Table with Top N filter
- Revenue by Gender — Bar chart (Male \$842K · Female \$673K · Other \$448K)
- Year-Over-Year Revenue — Clustered bar (2023 \$926K vs 2024 \$1.04M · +32% growth)
- Slicers: Category · Age Group · Season · KMeans Label

6.2 Page 2 — Customer Intelligence

- 4 Segment KPI Cards: Champions 246 · Loyal 1,293 · At-Risk 1,100 · Lost 1,111
- RFM Cluster Scatter Plot — Recency vs Monetary colored by KMeans segment
- Customers per Segment — Horizontal bar chart
- Segment Distribution — Donut chart with % labels
- Top 5 Customers Table — Revenue + RFM Scores + R/F/M breakdown
- Average RFM Score by Segment — Bar chart
- Avg CLV by Segment — Champion \$2,084 · At-Risk \$1,815 · Loyal \$1,682 · Lost \$267
- Slicers: Segment · Age Group · Gender · KMeans Label · CLV Tier

6.3 Page 3 — Sales Analysis

- 5 KPI Cards: Total Revenue · Total Orders 5K · AOV \$392.62 · Top Category (Clothing) · Top Season (Fall)
- Monthly Revenue Trend 2023 vs 2024 — Line chart
- Revenue by Category — Horizontal bar chart
- Category × Season Heatmap — Matrix visual with conditional formatting
- Revenue by Payment Methods — Horizontal bar (Debit \$477K · Cash \$453K · Credit \$433K)
- Top 5 Items — Bar chart by count
- Slicers: Date · Category · Season · Discount Applied · Payment Method

6.4 Page 4 — Churn Analysis

- 5 KPI Cards: Churn Rate 42.72% · Churned 2,136 · Active 2,864 · Avg Churn Prob 0.43 · Revenue at Risk \$624.9K
- Churn Rate vs Target — Gauge chart (current 42.72% vs target 27%)
- Churn Rate by Segment — Bar (Lost 72.7% · Potential 49.6% · At-Risk 44.8% · Loyal 16.1% · Champion 13%)
- Churn by Age Group — Bar (26-35 highest at 44.4%)
- Active vs Churned — Donut chart (57.3% vs 42.7%)
- Win-Back List — Top 20 At-Risk customers with Churn Probability + Recommended Action
- Churn by Subscription Status — Bar (No-Sub 54.3% vs Subscribed 17.5%)
- Slicers: Segment · Age Group · Gender · Subscription · CLV Tier

6.5 Page 5 — CLV & Loyalty

- 5 KPI Cards: Total Predicted CLV \$5.18M · Avg CLV \$1,035 · High CLV 1,700 · Low CLV 1,650 · Champion CLV \$2,084
- Customer Loyalty Funnel — All Customers 5,000 → Active 2,864 → Repeat 1,293 → Loyal 1,100 → Champion 246
- Avg CLV by Segment — Horizontal bar chart
- CLV Tier Distribution — Donut (High 34% · Medium 33% · Low 33%)
- CLV by Subscription — Yes \$1,307 vs No \$910 (43.6% higher)
- Avg CLV by Age Group & Gender — Combined bar + donut
- Slicers: CLV Tier · Age Group · Gender · Subscription · Segment

7. ML & Analytics Methodology

Component	Method	Tool	Output / Result
RFM Scoring	Quintile-based 1–5 scoring	Python · pandas	6 segments: Champion, Loyal, Potential, New Customer, At-Risk, Lost
Segmentation	K-Means Clustering (K=5)	sklearn · PCA	5 KMeans clusters with Silhouette scoring + PCA 2D visualization
CLV Prediction	Gradient Boosting Regressor	sklearn · 4 models	R ² score · MAE · CLV Tiers: High / Medium / Low per customer
Churn Model	Gradient Boosting Classifier	sklearn · 4 models	AUC-ROC target 80%+ · Churn Probability 0–1 · Win-back list
SQL Layer	Window Functions + CTEs	MySQL Workbench	5 scripts · vw_kpi_summary view · Power BI ready
Dashboard	DAX Measures + Slicers	Power BI Desktop	5-page interactive report · drill-through · slicer sync

8. Key Findings & Insights

8.1 Revenue Insights

- Total revenue of \$1,963,079 across 5,000 customers over Jan 2023 – Dec 2024
- Clothing is the top category at \$776,802 (39.6% revenue share)
- Fall is the peak season generating \$559,693 (28.5% of total revenue)
- 2024 revenue grew 32% vs 2023 (\$1.04M vs \$926K)
- Debit Card is the #1 payment method at \$476,867 (24.3%)
- Electronics has the highest avg CLV at \$3,045 per customer despite lower order volume

8.2 Customer Segmentation Insights

Segment	Count	Avg CLV	Total Revenue	Churn Rate	Avg Rating	Priority
Champion	300	\$2,084	\$~150K	13.0%	4.1	Protect
Loyal	1,159	\$1,682	\$630K	16.1%	3.8	Nurture
Potential	1,057	\$563	\$480K	49.6%	3.6	Upgrade
New Customer	784	\$571	\$310K	44.5%	3.5	Onboard
At-Risk	714	\$1,815	\$463K	44.8%	3.4	Urgent
Lost	986	\$267	\$147K	72.7%	3.2	Win-Back

8.3 Churn Analysis Insights

- Overall churn rate: 42.7% (2,136 customers) — 15.7% above the 27% BRD target
- Revenue at risk from churned customers: \$624,912
- 419 High CLV customers have churned — \$347,766 revenue at risk (priority win-back)
- Non-subscribers churn at 54.3% vs subscribers at 17.5% — 36.8% reduction through subscription
- 26-35 age group has highest churn rate at 44.4%
- Champion segment has lowest churn at only 13% — highest value, most loyal
- Lost segment has highest churn at 72.7% — immediate win-back campaign needed

8.4 CLV Insights

- Total predicted CLV across all 5,000 customers: \$5,176,565
- Champion segment: highest avg CLV at \$2,084 — protect at all costs
- At-Risk segment: second highest avg CLV at \$1,815 — urgent retention needed
- Electronics customers have highest category CLV at \$3,045 per customer
- Subscribed customers have 43.6% higher CLV (\$1,307 vs \$910) than non-subscribers
- 18-25 age group has highest avg CLV at \$1,114 — highest lifetime potential

9. Recommendations

#	Action	Details	Impact	Priority
01	Win-Back At-Risk Customers	Personal outreach + discount for 419 High CLV churned customers (\$347K revenue at risk). Email campaigns for full 1,100 At-Risk segment.	Revenue +\$347K	CRITICAL
02	Convert to Subscriptions	Push subscription campaigns to 3,421 non-subscribers. Subscription reduces churn by 36.8%. Target: convert 1,000 customers → save ~367 churns/year.	Churn -36.8%	HIGH
03	Electronics Upsell Program	Electronics CLV \$3,045 is highest category but only 370 orders. Targeted upsell and cross-sell campaigns in this category.	Revenue +\$420K	HIGH
04	Protect Champion Customers	246 Champions with 13% churn generate 40%+ revenue. VIP program, exclusive offers, priority service to retain this segment.	CLV +\$500K	HIGH
05	Fall Season Investment	Fall drives \$560K (28.5%). Pre-season inventory planning, targeted promotions, and campaign budget allocation before Fall.	Revenue +15%	MEDIUM
06	Service Recovery Program	882 customers rated below 3.0 — high churn probability. Follow-up calls and service recovery to improve ratings and reduce churn risk.	Retention +8%	MEDIUM

10. Non-Functional Requirements

Requirement	Specification
Dashboard Load Time	< 5 seconds for all 5 pages
UI Design	Clean, consistent dark theme matching CBIA dashboard colors
Interactivity	Full filtering, slicers, and drill-through on all 5 pages
Data Accuracy	100% KPI accuracy validated against source churn_predictions.csv
Churn Model Accuracy	Minimum 80% AUC-ROC score (achieved: model validated)
CLV Model Accuracy	R ² score validated against test set
File Formats	Power BI .pbix · Python .ipynb · SQL .sql · Excel .xlsx · PPT .pptx
GitHub Repository	Public repo with README, folder structure, and all deliverables
SQL Compatibility	MySQL Workbench 8.0+ compatible scripts

11. Project Deliverables

#	Deliverable	File Name	Format	Phase
1	Business Requirements Document	CBIA_BRD_v2.docx	.docx	Phase 0
2	Customer Dataset (Cleaned)	customer_cleaned.csv	.csv	Phase 1
3	EDA & Cleaning Notebook	01_EDA_Data_Cleaning.ipynb	.ipynb	Phase 2
4	RFM Scoring Notebook	02_RFM_Scoring.ipynb + rfm_scored.csv	.ipynb/.csv	Phase 3
5	KMeans Clustering Notebook	03_KMeans_Clustering.ipynb	.ipynb	Phase 4
6	CLV Prediction Notebook	04_CLV_Prediction.ipynb	.ipynb	Phase 5
7	Churn Model Notebook	05_Churn_Model.ipynb	.ipynb	Phase 6
8	SQL Analytics Scripts	06_SQL_Scripts_MySQL.sql	.sql	Phase 7
9	Power BI Dashboard	dashboard.pbix	.pbix	Phase 8
10	Project Presentation	CBIA_Presentation_v2.pptx	.pptx	Phase 9
11	GitHub Repository	github.com/deepak/CBIA	Public	Phase 9

12. Acceptance Criteria

- Dashboard accurately reflects all defined KPIs with correct calculations validated against churn_predictions.csv
- All 5 Power BI pages render correctly with slicers, filters, and drill-through functioning
- RFM segmentation logic produces 6 segments: Champion, Loyal, Potential, New Customer, At-Risk, Lost
- Churn model AUC-ROC score meets minimum 80% threshold
- CLV predictions validated against actual revenue with reasonable R² score
- SQL scripts return results consistent with Python notebook outputs
- All 5 Power BI pages respond to slicer selections within 5 seconds
- Win-back list correctly identifies churned High CLV customers with recommended actions
- GitHub repository contains all 11 deliverables with clear README documentation
- Presentation deck (10 slides) covers all project phases with accurate data points

13. Risks & Mitigation

Risk	Impact	Mitigation Strategy
Data Quality Issues	High	Data cleaning, null handling (90 Age_Group nulls fixed), outlier removal in EDA notebook
Model Accuracy Below Target	Medium	Cross-validation, 4-model comparison, hyperparameter tuning, feature engineering
Power BI Performance	Low	Data model optimization, aggregation tables, direct import from clean CSV
KPI Misinterpretation	Medium	All numbers verified against source data; BRD documents exact calculation methodology
MySQL Import Errors	Medium	Renamed problem columns (Qty, Purchase_Amount_USD); syntax converted from SQLite to MySQL

14. Approval & Sign-Off

Role	Name	Signature	Date
Project Author / Analyst	Deepak Kumar Prasad	_____	May 2026
Client Representative	XYZ Pvt. Ltd.	_____	May 2026
Reviewer	_____	_____	_____

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